PDP Paper-to-Notebook Reader Guide

A navigation companion for *A Paired-Control Architecture for QRNG Experiments: Operating Characteristics and Inferential Boundaries* (8-18-26 edition)

Purpose. This guide lets a reader move from a statement in the paper to the notebook analysis that produced or tested it. Notebook locations are identified by visible markdown headers, not cell or line numbers.

What each entry provides

- Paper section and claim
- Notebook name and visible header
- What question the analysis addresses
- How the analysis was performed
- The result produced by the executed analysis
- Dataset, population, and inferential unit
- Whether the notebook is the primary result source or a secondary sensitivity check

Important scope distinction. Pilot 4 results are produced primarily by NB1 and NB2. The Interleaving notebook is a Pilot 4 retrospective reconstruction used for partitioning sensitivity and limitations. The Exp4-versus-Exp5 notebook is a secondary cross-dataset check. The separate 80-block local-coupling dataset is outside Pilot 4 and is available from the author on request.

Notebook index

NB1	Exp4_Notebook1_Zen_Paired_Control_Stream_Design_Validation (3).ipynb	Design validation, PCS calibration, Baseline relational behavior, injection checks, QRNG audit reconstruction, and synthetic Baseline-offset checks.
NB2	Exp4_Notebook2_Zen_Temporal_Structure_Analysis (2).ipynb	Hit-rate results, single-scale H_{RS} analyses, subgroup and threshold diagnostics, temporal sensitivity, local and lagged coupling, and finite-sample calibration.
Interleaving	Exp4_Interleaving_PhysicalHalf_AI_and_StreamLength_Audit.ipynb	Retrospective reconstruction of contiguous versus locally balanced partitions, deterministic odd-even allocation, AI comparison, and stream-length variance scan.
Exp4 vs Exp5	Exp4_vs_Exp5Prescreen_Baseline_CrossDataset_Check.ipynb	Secondary comparison of Pilot 4 Baseline behavior with a dated Exp5-prescreen snapshot. Not a Pilot 4 primary inferential notebook.

How to find an analysis

1. Find the paper section in this guide. 2. Open the named notebook. 3. Use the notebook search function to locate the visible header exactly or by its distinctive phrase. 4. Read the markdown immediately above the code for the estimand and limitations. 5. Read the saved output immediately below the code for the reported result.

> When more than one notebook is listed, the first is the primary source unless the Source role field says otherwise. A secondary notebook may use the same Pilot 4 estimate for a different diagnostic question.

3.1 / Figure 3

PCS calibration and condition comparisons

Notebook and header	NB1 - PCS Stream Calibration; PCS Randomness: Formal Mean Tests and Effect Sizes; PCS Distribution Shape
What it does	Uses all retained Pilot 4 PCS blocks. It checks the overall PCS hit rate, session-clustered uncertainty, and whether PCS hit-rate or single-scale H_{RS} summaries differ across Human, AI, and Baseline.
How it was done	Calculated the PCS target-aligned hit-rate mean over retained blocks; obtained uncertainty by resampling complete sessions; tested the mean against .50 with session-clustered sign flips. Condition comparisons used session-level summaries and one-way ANOVA/permutation companions for PCS hit rate and PCS H_{RS}.
Result shown	Overall PCS hit rate = 0.500253, 95% CI [0.499504, 0.501022], sign-flip p=.5160. No condition difference was detected for mean PCS hit rate (p=.7769) or mean PCS H_{RS} score (p=.4696).
Population / data	11,607 retained blocks; sessions preserved for inference
Source role	Primary result source

3.2.1

Baseline same-call Subject-PCS correlation

Notebook and header	NB1 - Analysis 1 - Same-call association diagnostic
What it does	Filters to Baseline and measures Pearson and Spearman association between the two simultaneous stream scores. Session bootstrap and within-session re-pairing assess uncertainty without treating blocks as independent.
How it was done	Filtered to Baseline. For each same-call pair, correlated Subject and PCS H_{RS} scores and target-aligned hit rates. Pearson uncertainty used complete-session bootstrap; permutation tests re-paired or permuted PCS within sessions; Spearman supplied a rank-based companion.
Result shown	H_{RS}: Pearson r=+0.0037, 95% CI [-0.0306,+0.0371], permutation p=.8357; Spearman rho=+0.0027, p=.8791. Hit rate: Pearson r=+0.0149, 95% CI [-0.0248,+0.0551], permutation p=.3966. No tested same-call association resolved.
Population / data	3,090 Baseline blocks; 103 sessions
Source role	Primary result source

3.2.2-3.2.4

Baseline paired-difference behavior, physical halves, and re-pairing

Notebook and header	NB1 - Analysis 2 - Paired-delta location and shape; Empirical physical-half calibration for Paper §3.3.4; Analysis 3 - Covariance contribution to paired-delta variance
What it does	Checks whether Baseline Subject-minus-PCS differences are centered near zero, whether physical halves A and B differ before labeling, and whether same-call pairing changes variance relative to re-paired calls.
How it was done	Computed block-level Subject-minus-PCS differences for H_{RS} and hit rate, then resampled complete Baseline sessions. Reconstructed physical halves A and B before the random label assignment. Compared observed same-call variance/covariance with within-session cross-call re-pairings that preserved marginal distributions and session membership.
Result shown	The Baseline paired H_{RS} mean was -0.001741 and its whole-session interval included zero. Physical A-minus-B H_{RS} was -0.000682 with interval [-0.003060,+0.001687]. Same-call covariance and re-pairing diagnostics did not show a resolved dependence at the tested resolution.
Population / data	3,090 Baseline calls; 103 sessions
Source role	Primary result source

3.2.5

Additional dependence and joint-stream diagnostics

Notebook and header	NB1 - Same-Position Bit-Level Coupling Diagnostics; Condition Comparisons of Subject-PCS Relationship Structure; Alternative Ordering Metrics Across All Conditions
What it does	Examines mutual information, correlations, variance ratios, energy distance, lagged structure, entropy, and alternative ordering metrics across conditions. It is a broad exploratory diagnostic battery.
How it was done	Ran a broad exploratory battery across conditions: same-position mutual information relative to circular shifts, correlations, variance comparisons, energy distance, lagged measures, entropy, and alternative ordering metrics. Human resampling preserved contributors and sessions; automated conditions preserved sessions or collection units where implemented.
Result shown	Same-position excess mutual information and the principal correlation/energy comparisons did not resolve clear condition differences. The saved energy-distance permutation p-values were .2082, .4020, and .6688. Some manuscript summaries of this large battery remain flagged for exact result reconciliation in the detailed audit.
Population / data	All Pilot 4 conditions; Human contributor hierarchy and session structure retained where implemented
Source role	Primary result source; several manuscript narrative summaries still require reconciliation

3.3

Controlled artifact-injection behavior

Notebook and header	NB1 - Upstream Artifact-Injection Validation and its shared-bias, persistence, Subject-only, and unequal-half subsections
What it does	Starts from retained Baseline raw calls, applies specified synthetic disturbances before assignment/scoring, and measures how much appears in Subject, PCS, and their paired difference. These are implementation and operating-characteristic tests, not naturally occurring artifacts.
How it was done	Started from retained Baseline raw calls and modified bits before stream labeling and scoring. Shared literal-bit and persistence disturbances were applied to both physical halves; Subject-only and unequal-half versions served as controls. Repeated realizations and session resampling quantified recovery and variability.
Result shown	Equal shared disturbances produced little systematic displacement in the paired difference, while uncertainty increased with dose. Subject-only target replacement produced paired hit-rate changes of approximately .0100, .0249, .0500, and .1000 at replacement probabilities .02, .05, .10, and .20. Unequal-half persistence left a small residual, demonstrating the cancellation boundary.
Population / data	3,090 retained Baseline raw calls; repeated simulations
Source role	Primary result source

3.4 / Table 2

Audit availability, reconstruction, pass rates, and condition comparison

Notebook and header	NB1 - Scheduled-Audit Availability and Missingness; Three-Subtest Audit-Battery Reconstruction and Session-Aware Summary; Audit-Failure Session Uniqueness - Stored-Only and Reconstructed
What it does	Reconstructs the three implemented audit subtests from the stored 1,000-bit sequences, verifies pass/fail classifications against 1,771 stored-complete audits, fills 120 missing subtest records, and analyzes all 1,891 audits with session-aware uncertainty and permutation comparisons.
How it was done	Recomputed Frequency, Runs, and Longest Run of Ones from each 1,000-bit audit. Validated reconstructed pass/fail outcomes against 1,771 audits with stored complete results, reconstructed 120 missing records, bootstrapped complete sessions, and compared condition failure rates through whole-session permutations with Holm adjustment.
Result shown	All three reconstructed classifications had zero pass/fail disagreements on the stored-complete subset. Of 1,891 audits, 1,845 passed and 46 failed, for 97.57% [96.89%,98.20%]. Failures occurred in 46 sessions. All Holm-adjusted condition-comparison p-values were 1.00.
Population / data	1,891 audits across 389 sessions
Source role	Primary result source; paper Table 2 Overall Audits should be 1,891

3.5 / Table 3

Mean-level target-aligned hit-rate results

Notebook and header	NB2 - Authoritative Four-Estimand Component Table; Population and AI Paired Hit-Rate Comparisons; Bayesian paired hit-rate model
What it does	Reports Subject and PCS component means, within-call Subject-minus-PCS differences, and contrasts with Baseline. Frequentist resampling preserves contributors and sessions; the Bayesian model uses the continuous block-level paired difference.
How it was done	Calculated Subject and PCS target-aligned means, within-call Subject-minus-PCS differences, and each condition contrast with Baseline. Frequentist intervals resampled Human contributors then sessions and resampled Baseline/AI collection units as specified. The Bayesian companion modeled the continuous block-level paired difference with identifier and session random intercepts and a Student-t likelihood.
Result shown	All Table 3 paired-difference and Baseline-contrast intervals included zero. The executed Bayesian replacement estimated paired differences near zero: Baseline +0.000184, AI +0.000193, Human 5+ +0.000601; the Human 5+-minus-AI contrast was +0.000407, with every 95% HDI including zero.
Population / data	Baseline, AI, all Human, and descriptive Human 5+ subgroup
Source role	Primary result source

3.6.1

Full-Human single-scale H_{RS} result

Notebook and header	NB2 - Single-Scale Hurst - Population-Level Null
What it does	Compares the full Human paired H_{RS} distribution and mean with Baseline. It includes hierarchy-aware uncertainty plus unclustered t and KS tests as descriptive sensitivity checks.
How it was done	Computed the single-scale H_{RS} score separately for Subject and PCS, formed the paired difference, and compared all Human blocks with Baseline. The principal interval preserved Human contributor-to-session clustering and Baseline sessions; unclustered Welch and KS tests were retained only as sensitivity descriptions.
Result shown	Human paired mean = +0.00113; Baseline = -0.00174; contrast = +0.00287 with an interval narrowly including zero. Unclustered Welch t=1.94, p=.0521; KS D=.0226, p=.2884. The full-cohort result was unresolved and method-sensitive.
Population / data	All Human versus Baseline
Source role	Primary result source

3.6.2 / Table 4 / Figure 4

Session-count pattern and Baseline offset interpretation

Notebook and header	NB2 - Session-Count Pattern (Descriptive) - Mean Paired ΔH_{RS} by Condition; NB1 - Baseline Offset - Synthetic-Pipeline Check and Bootstrap Seed Stability; Exp4 vs Exp5 - Statistic 1 - Within-Baseline paired delta
What it does	The new NB2 table is the direct source for Paper Table 4. It describes mean paired single-scale H_{RS} differences for mutually exclusive Human session-count strata and supplies Baseline, AI, and pooled Human context rows. NB1 tests whether the pipeline itself creates a stable Baseline offset. The cross-dataset notebook compares the realized Pilot 4 Baseline value with an Exp5-prescreen snapshot.
How it was done	For every retained block, calculated ΔH_{RS} as Subject H_{RS} minus PCS H_{RS}, then took block-weighted means within condition and Human session-count stratum. Each Human contributor was assigned once according to total completed sessions, so the 1, 2-4, and 5+ groups do not overlap. No significance test is attached to this descriptive table. The contextual Baseline, AI, and pooled-Human rows are shown in NB2 but are not all printed as rows in Paper Table 4. NB1 separately ran Bernoulli(.5) 301-bit calls through the unchanged pipeline; the cross-dataset notebook compared Pilot 4 with an Exp5-prescreen snapshot.
Result shown	Paper Table 4 reports: exactly 1 session, 3,327 blocks, mean ΔH_{RS}=-0.00004; 2-4 sessions, 570 blocks, +0.00322; 5+ sessions, 840 blocks, +0.00435. These are descriptive recruitment-group summaries, not evidence that completing more sessions caused the pattern. Baseline mean was -0.001741 in Pilot 4 and -0.000459 in the Exp5-prescreen snapshot; both intervals included zero, and the synthetic-pipeline check did not produce a consistent offset.
Population / data	Pilot 4 Baseline, AI, pooled Human, and mutually exclusive Human strata: exactly 1 session, 2-4 sessions, and 5+ sessions; secondary Exp5-prescreen Baseline snapshot
Source role	NB2/NB1 primary; Exp4-vs-Exp5 secondary sensitivity source

3.6.3 / S3 / Table S1 / Figure S1

Shuffle sensitivity of the H_{RS} score

Notebook and header	NB2 - Single-Scale Hurst Delta - Shuffle Test
What it does	Applies the same within-window positional permutation to Subject and PCS within each paired block, preserving composition and pairing while changing order. It reports mean Human 5+-versus-Baseline KS distances across window sizes.
How it was done	For each paired block, applied the same positional permutation to Subject and PCS within nonoverlapping windows W=3,5,10,25,50,150. This preserved each stream's composition, pairing, and shared positional transformation. Repeated each nontrivial window 200 times and recalculated the Human 5+-versus-Baseline KS distance.
Result shown	Unshuffled KS distance = .0541. Mean shuffled distances were .0491, .0600, .0628, .0586, .0395, and .0333 from W=3 through W=150. Separation changed with ordering and was smallest after full-block shuffling; this was descriptive, not participant-level inference.
Population / data	Human 5+ and Baseline blocks; 200 realizations per nontrivial window
Source role	Primary result source

3.6.4 / S4

Direct versus paired interpretation, stream decomposition, thresholds, and label swap

Notebook and header	NB2 - Single-Scale Hurst - Power Users (5+ Sessions); Authoritative Four-Estimand Component Table; composition-conditional threshold calibration; label-swap sensitivity
What it does	Separates the direct Subject-stream comparison from the paired Subject-minus-PCS comparison, shows how Subject and PCS directions combine, evaluates nested 2+/3+/5+ thresholds, and performs assignment-label sensitivity checks.
How it was done	Compared Human 5+ and Baseline in four linked views: direct Subject-stream H_{RS}, paired Subject-minus-PCS H_{RS}, separate Subject/PCS component directions, and nested 2+/3+/5+ threshold selection. Label-swap permutations tested assignment sensitivity; composition-conditional simulations randomized bit order while preserving stream composition.
Result shown	Direct KS p=.0243 and paired KS p=.0398 were nominal block-level results from the same data, not independent confirmation. Subject shifted upward and PCS downward. The fixed 5+ random-order tail area was .00695; the maximum across 2+/3+/5+ thresholds was .00800. This correction did not cover the wider exploratory program.
Population / data	Human session-count groups, Human 5+, AI, and Baseline
Source role	Primary result source

3.6.5-3.6.7

Run structure, composition, entropy, and variance checks

Notebook and header	NB2 - Run-Structure Check - Human 5+ vs Baseline; bit-composition and entropy diagnostics; paired-H_{RS} variance comparison
What it does	Tests whether simpler run, composition, entropy, or variance differences reproduce the candidate H_{RS} contrast. These diagnostics help characterize the metric but do not independently confirm the subgroup result.
How it was done	Calculated run counts, maximum runs, run-length entropy, bit composition, n-gram entropy, and paired-H_{RS} variance. Repeated Human 5+-versus-Baseline contrasts using hierarchy-preserving resampling and the same paired construction where applicable.
Result shown	Simpler run, composition, and entropy measures did not reproduce a resolved Human 5+-Baseline contrast. Paired-H_{RS} variance was 0.004091 in Human 5+ and 0.004116 in Baseline, ratio .994 with interval [.865,1.172]. No increased subgroup variance was detected.
Population / data	Human 5+ versus Baseline, with broader condition summaries where shown
Source role	Primary result source

3.7.1 / S1.1

Within-session position and early/late sensitivity

Notebook and header	NB2 - Within-Session Position Sensitivity Analysis; Within-Session Hurst-Delta Drift - Human 5+ vs Baseline Interaction Test; Interleaving - 2. Original vs interleaved Baseline summary
What it does	NB2 measures continuous session-position slopes, early/late means, and the Human 5+-by-half interaction. The Interleaving notebook independently reconstructs the original contiguous split and shows how the Baseline early/late pattern changes when positions are redistributed.
How it was done	Estimated continuous within-session H_{RS}-delta slopes after chronological sorting. A post-hoc split divided blocks 0-14 from 15-29. The direct Human 5+-by-half interaction resampled Human contributors then sessions and Baseline sessions. The Interleaving notebook reconstructed the same Baseline calls under alternative position allocations.
Result shown	All continuous slope intervals included zero. Human 5+ early and late means were +.005857 and +.002842; Human 5+-Baseline contrasts were +.009921 early and +.002259 late. Late-minus-early interaction = -.007662, 95% CI [-.019489,+.004125], unresolved. Interleaving attenuated the Baseline early/late pattern.
Population / data	Pilot 4 complete sessions; Human 5+ contains 3 contributors and 28 sessions; Baseline has 103 sessions
Source role	NB2 primary result; Interleaving secondary mechanism/sensitivity source

S1.2-S1.4

Weighting, recurrence, PI influence, and session order

Notebook and header	NB2 - Hierarchy-Preserving Bootstrap Sensitivity; Human 5+ Session Recurrence and PI-Influence Check; First Session Versus Later Sessions Among Eventual Human 5+ Contributors
What it does	Checks whether the subgroup result depends on block versus contributor weighting, one session, the principal investigator, or first-versus-later session ordering.
How it was done	Recomputed the Human 5+ contrast under block, equal-session, and equal-contributor/equal-session weighting; bootstrapped contributors then sessions; removed each session and the principal investigator in turn; compared each contributor's first session with later sessions.
Result shown	Block and equal-session contrast = +.00609; equal-contributor/equal-session = +.00818. Leave-one-session-out paired means ranged +.003270 to +.005046. Excluding the PI increased the contrast to +.010423. Later-minus-first changes were mixed across contributors.
Population / data	Three Human 5+ contributors and their 28 sessions; Baseline sessions for contrasts
Source role	Primary sensitivity source

3.7.2 / S2

Local and lagged Subject-PCS coupling

Notebook and header	NB2 - Exploratory Relational Analysis - Local Coupling Stability; Exploratory Relational Analysis - Lagged Cross-Correlation Family
What it does	Calculates correlations inside nonoverlapping session windows and cross-correlations at lags -3 through +3. Mean association and variability are separate estimands, and the lag analysis applies a 14-comparison family correction.
How it was done	Within each complete session, calculated Pearson correlations in nonoverlapping windows of 5 and 10 blocks and compared their means and variances. Separately calculated session-level Subject(t)-PCS(t+k) correlations at lags -3 to +3. Seven mean plus seven variance comparisons formed a 14-cell family.
Result shown	Human 5+ window-correlation variance ratios were .832 at W=5 and .742 at W=10, but only three contributors prevented reliable population intervals. At lag +2, all-Human variance ratio = .5505, Levene p=.00110, sensitivity interval [.383,.808]; mean difference +.0354, p=.154. The lag variance result survived the stated family corrections but remains exploratory.
Population / data	Human, Human 5+, AI, and Baseline as specified by each comparison
Source role	Primary Pilot 4 source; the 80-block external dataset is available separately on request

4.2, 4.4, 4.7, and 5.11

Interpretation and limitation of contiguous stream partitioning

Notebook and header	Interleaving - 1. Reconstruct original and PRNG-interleaved streams; 2. Original vs interleaved Baseline summary; 6. Reconstruction-null test; Deterministic odd/even check; 7. AI condition; 8. Baseline stream-length variance scan
What it does	Reconstructs Pilot 4 raw calls under the original contiguous split and alternative locally balanced partitions. It compares early/late behavior across 500 pseudorandom reconstructions, performs a deterministic odd-even check, checks AI, and explores how H_{RS} variability changes with stream length.
How it was done	Reconstructed every retained Pilot 4 Baseline call under the original contiguous split and locally balanced pseudorandom partitions, repeated the reconstruction-null comparison 500 times, ran a deterministic odd-even allocation, repeated the early/late check in AI, and scanned H_{RS} variability over alternative stream lengths.
Result shown	Original Baseline early-minus-late = -.004647; one interleaved reconstruction = +.001362. Across 500 reconstructions the median was -.000163 with 95% range [-.004166,+.004026]; empirical two-sided tail area for the original contrast = .03. Deterministic odd-even gave +.001418. The position pattern was attenuated, but the retrospective test does not establish causation or general superiority.
Population / data	Pilot 4 Baseline raw calls; AI for the condition check
Source role	Primary source for the interleaving discussion and contiguous-partition limitation

5.10

Finite-sample behavior of the single-scale H_{RS} score

Notebook and header	NB2 - bounded AR(1)/fractional-Gaussian-noise calibration and directional-asymmetry checks
What it does	Simulates the exact 150-bit statistic under bounded generating models and observed composition margins. It evaluates response to imposed ordering differences and whether forward-versus-reverse estimator asymmetry could account for the observed contrast.
How it was done	Generated 150-bit rank-binarized Gaussian AR(1) and fractional-Gaussian-noise sequences across bounded parameter pairs, reproduced observed Human 5+ composition margins, calculated the exact implemented H_{RS} statistic, and compared forward and reverse assignment directions.
Result shown	The score followed imposed ordering contrasts in the expected direction. Every directional-asymmetry interval included zero; the largest endpoint was about .00152 versus the observed Human 5+ paired difference +.00435. Within the simulated cases, estimator asymmetry did not account for the full observed contrast.
Population / data	Synthetic 150-bit sequences calibrated to the Human 5+ composition margins
Source role	Primary calibration source

Cross-dataset sensitivity context

Pilot 4 versus Exp5-prescreen Baseline comparison

Notebook and header	Exp4 vs Exp5 - Statistic 1 - Within-Baseline paired delta; Statistic 2 - Baseline level vs Human PCS level
What it does	Reconstructs the Pilot 4 Baseline from frozen raw calls and compares it with a dated, ongoing Exp5-prescreen snapshot. It examines within-Baseline paired H_{RS} offsets and whether Baseline levels differ from Human PCS levels. It is not part of the Pilot 4 primary inferential dataset.
How it was done	Reconstructed Pilot 4 Baseline from frozen raw calls and processed the dated Exp5-prescreen snapshot with the same paired H_{RS} definition. Bootstrapped Pilot 4 sessions and Exp5 sessions. Also compared each dataset's Baseline level with its Human PCS level under naive and deduplicated contributor definitions.
Result shown	Within-Baseline paired means were -.001741, 95% CI [-.003811,+.000355], for Pilot 4 and -.000459 [-.002286,+.001408] for Exp5-prescreen. Baseline-minus-Human-PCS intervals included zero in both datasets and under all Exp5 deduplication variants.
Population / data	Pilot 4 Baseline and dated Exp5-prescreen snapshot
Source role	Secondary cross-dataset sensitivity source

Known navigation and reconciliation notes

Separate 80-block dataset

The S2.1 longer-window local-coupling statement concerns a dataset outside Pilot 4. It is not expected in NB1, NB2, or the Pilot 4 interleaving notebook. Supporting materials are available from the author on request.

Cross-dataset notebook

Its Exp5-prescreen values are not primary Pilot 4 results. It is a secondary sensitivity source for interpreting whether the negative realized Pilot 4 Baseline mean appears to be a general processing offset.

Interleaving notebook

This notebook is the direct source for the paper's statements about attenuation under locally balanced interleaving, the 500 reconstruction-null repetitions, the deterministic odd-even check, and the contiguous-partition limitation.